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"They told me to succeed where they couldn't": Parent support and STEM motivation among STEM majors from low socioeconomic status backgrounds

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Abstract
Grounded in situated expectancy-value theory, this retrospective mixed-methods study examined how parent support during adolescence relates to STEM (science, technology, engineering, and math) motivation among low SES undergraduates currently majoring in STEM ($n=222$; 86% women; 53% White, 19% Latine, 17% Asian, 7% Black). Past work has demonstrated that parent support positively relates to STEM motivation. However, few studies have examined this pattern amongst low socioeconomic status (SES) students, defined as students who are first-generation and/or from low-income backgrounds. Open-response thematic analysis revealed that socioemotional support, instrumental support, and role models were the most common forms of support for adolescent STEM pathways. Regression analyses revealed that quantitative measures of socioemotional support predicted student's STEM expectancy and value beliefs, while instrumental support predicted only STEM expectancy beliefs. The implications of such findings can assist educators and parents in creating family-centered outreach opportunities that emphasize parental values and encouragement to help broaden STEM participation among low SES youth.

Keywords First-generation · Family support · Low-income · Expectancy-value beliefs

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1 Introduction

Despite national efforts to diversify the pathway of students into the STEM workforce, students from low socioeconomic status (SES) backgrounds remain underrepresented in STEM majors (Bettencourt et al., 2020; Garriott & Nisle, 2018; Jiang et al., 2020) and careers (National Science Board, 2024). Similar gaps have been observed earlier in development, where low SES high school students report lower STEM careers aspirations compared to their higher SES peers (Gao & Eccles, 2020; Garriott et al., 2013; Starr et al., 2022b). Research has documented that parents and families play an integral role in STEM pathways for Black and Latine students (Starr et al., 2025; Onuma et al., 2022; Richardson et al., 2020; Rodriguez et al., 2019; Russell & Atwater, 2005). However, few quantitative or qualitative studies exist that focus on the role of parents and family in promoting STEM futures among low SES adolescents. Drawing from expectancy-value theory we conceptualize the role of parents as key socializers of motivational beliefs in STEM.

The present retrospective mixed-methods study examined how STEM undergraduates from low SES backgrounds recall their families influencing their STEM motivation during adolescence. We defined "low SES" as students who identified as coming from a low-income background and/or being the first in their family to attend college. We chose current undergraduate STEM majors to highlight low SES students who have been successful in their STEM aspirations thus far. By asking them to report on how their families influenced or supported them in STEM as adolescents, both qualitatively (through open response) and quantitatively (through hierarchical regression), we can learn more about how STEM support in adolescence may support successful STEM motivation and persistence in college.

1.1 Situated expectancy-value theory and social class

Situated expectancy-value theory (Eccles & Wigfield, 2020) provides a framework to examine why some students may persist in STEM whereas others do not. The theory posits that students' achievement-related performance (such as test scores) and choices (such as course enrollment) are strongly influenced by their motivational beliefs in a domain such as STEM (Eccles & Wigfield, 2020). That is, situated expectancy-value theory proposes that whether someone decides to pursue an academic domain depends on their motivational beliefs, including their expectancy beliefs (how confident one is in their ability) and value beliefs (Eccles & Wigfield, 2020). Value beliefs are made up of utility value (how useful one views a domain), interest value (how interesting one views a domain), attainment value (how personally valuable one views a domain), and cost (the perceived costs to pursuing a domain) (Eccles & Wigfield, 2020). STEM motivational beliefs may be particularly important to investigate during adolescence and emerging adulthood, as this is a developmental period when STEM motivation influences STEM college and career choices (e.g., Andersen & Ward, 2014).

Situated expectancy-value theory further highlights how motivational beliefs are situated and influenced by social contexts, including demographic characteristics such as parent education level and income (Eccles & Wigfield, 2020). Family social class

is strongly related to the quality of children's academic and STEM contexts (Eccles, 2005). Despite the desire to succeed academically and strong family connections, low SES students may not receive as much academic support as their more privileged peers due to barriers (Azmitia et al., 2018; Covarrubias et al., 2019; Covarrubias & Valle, 2023; Ives & Castillo-Montoya, 2020; Valle & Covarrubias, 2024; Sy et al., 2011; Watson & Azmitia, 2025). For example, low SES students may have parents with less academic/STEM capital and attend schools with fewer resources, limiting their access to quality STEM-related experiences (Archer et al., 2012; Aschbacher et al., 2010; Bettencourt et al., 2020; Martin et al., 2013; Talley & Ortiz, 2017). Furthermore, low SES students may experience social class discrimination in schools which affects their achievement and ability to learn (Gorski, 2012; see Zajda, 2022 for a review). Despite these barriers, parent support in STEM may protect low SES students from lower STEM motivational beliefs and help them persist in STEM. We discuss this in greater detail below.

1.2 STEM parent support among low SES families

Decades of developmental and educational psychology research indicate that families, in particular parents, play a central role in shaping the educational aspirations, confidence, and attainment of youth (e.g., Eccles & Wigfield, 2020), including in STEM fields (e.g., Bueno et al., 2022; Dotterer, 2022; Simpkins et al., 2018; Soto-Lara & Simpkins, 2020). Much of what is known about STEM-specific parent support is based on samples of middle-class White parents who may have greater access to academic and social capital, financial resources, and STEM-related opportunities (Archer et al., 2012; Starr et al., 2022b). As a result, there is comparatively little work examining the strengths low SES families offer to their adolescents' STEM development. However, several studies examining general STEM parent support among low SES (non-college educated and/or low-income) families suggest that parent support is positively correlated with STEM motivation (Jiang and Simpkins 2024; Starr et al., 2022b, 2026b; Wu et al., 2025). For example, Starr and colleagues (2026b) found that although declines were common in STEM motivational beliefs across high school among adolescents from non-college educated families, parent support was related to increases in motivational beliefs. Perceived parent support may be particularly important for the development of motivational beliefs among low SES adolescents given that they tend to have close family connections or a desire to help family, which are positive factors in college persistence (Covarrubias et al., 2019; Guiffrida, 2006; Stephens et al., 2012). Work that has been done among both high and lower SES families suggest that parents support their adolescents in STEM through instrumental support, socioemotional support, and role modeling.

Instrumental support refers to tangible support in STEM, including providing support via coactivity, advice giving, provision of materials/activities, or support on applications/homework (Simpkins et al., 2005). Prior work that does not specifically examine low SES families has found that instrumental support, such as providing STEM coactivities, generally relates to higher STEM motivational beliefs (see Hill & Tyson, 2009; Starr et al., 2022a for relevant meta-analyses). However, prior work has also suggested that unwanted or excessive instrumental support can reduce STEM

motivational beliefs amongst adolescents (Bhanot & Jovanovic, 2009; Shirefley, 2025). Furthermore, limited academic and STEM social capital may result in low SES parents providing less instrumental support related to STEM content for their children (Archer et al., 2012; Covarrubias et al., 2020; Sy et al., 2011). However, recent work has suggested that even when low SES parents provide less instrumental STEM support (compared to higher SES parents), their support is still strongly correlated to high STEM career aspirations among their adolescent children (Starr et al., 2022a, 2022b). This may be due to the idea that instrumental support among low SES youth may look different, relying less on content-expertise but physical helping such as assisting a child get to and from activities, or alleviating burdens like meals to leave more space for academic learning (Bueno et al., 2022).

Another way in which parents may support their adolescents in STEM is through socioemotional support, which may come in the form of encouraging an adolescent to pursue STEM courses or careers, noticing and complimenting an adolescent's STEM ability, or encouraging an adolescent to persist through a difficult STEM course (Starr et al., 2025). Research examining parental support of middle-class adolescents found that broad encouragement, emotional support, and high expectations predicted adolescents' motivation and achievement across several domains (Hill & Tyson, 2009; Shirefley, 2025). Although this work is more limited among low SES students, a prior study suggests that parental expectations are predictive of college going among adolescents in non-college educated families (Bui & Rush, 2016). Other work has shown that subcomponents of broader encouragement such as parents' affirmations of competence, expressions of pride, and messages related to persistence helped adolescents, including low SES adolescents, form positive ability beliefs and value academic domains, even when parents were unable to provide instrumental support (Bueno et al., 2022; Guiffrida, 2006; Packard et al., 2012; Pomerantz et al., 2006).

Prior work among Black and Latine families suggests that this parental socioemotional support via encouragement may be more predictive of motivational beliefs in STEM than instrumental support (Starr et al., 2025). Socioemotional parent support may be more strongly related to STEM motivational beliefs compared to instrumental support for two main reasons. First, encouragement may be more personally relevant and unambiguously positive. In contrast, instrumental support may be less child-directed. For example, a parent may insist on helping their child with their math homework or engaging in a science coactivity due to worrying about their child's performance, which could make their support unwanted or performance-contingent (DiStefano et al., 2023; Tulagan & Eccles, 2023). In contrast, many forms of socioemotional support, such as telling an adolescent that you are proud of their math abilities, are uniformly positive. Second, low SES students are often marginalized in school settings; for example, adolescents may experience discrimination from teachers and classmates and be stereotyped by teachers (e.g., Gorski, 2012; Mello, 2025; Starr et al., 2026a). Low SES parents' encouragement may counter this lack of encouragement and provide students with insights into their abilities and potential. Thus, although both instrumental and socioemotional parent support are important, socioemotional support may be more strongly related to STEM motivation among low SES students and families.

Parental STEM support may also take the form of conveying messages of cultural values, including perseverance and helping others. Additionally cultural values may communicate the importance of social mobility and financial stability (Auerbach, 2007; Guiffrida, 2006; Jackson & Suizzo, 2015). In several studies, researchers have documented that students from low SES backgrounds were more likely than their higher SES peers to choose pragmatic careers within STEM and other health-related fields due to the perceived social mobility and economic value of the careers (Aschbacher et al., 2010; Goyette & Mullen, 2006; Stanton-Salazar, 2011). Furthermore, students from low SES families may be motivated by the goal of helping others, including family members and people in their communities (Boekeloo et al., 2015). Many STEM fields, particularly health-related fields, are perceived as affording opportunities for helping others while also offering financial stability (Yang & Barth, 2015).

Finally, role models may also be impactful for adolescents, including low SES adolescents. Prior work has documented that for many adolescents, exposure to science and technical work frequently occurs through informal networks and social contacts, including parents and other family members in science-related or technical occupations. These connections can shape science aspirations and identity (Archer et al., 2012; Aschbacher et al., 2010; Delgado et al., 2021; Kuchynka et al., 2022). Although low SES students may be less likely to have role models in STEM due to less academic capital (Costello et al., 2023), prior work has documented that familial role models may be particularly important for students of marginalized backgrounds including low SES youth, as these models can offer concrete examples of how skills are used in everyday work (Goyette & Mullen, 2006) and provide an example of someone in STEM careers who is like them (e.g., similar race, SES, gender) (Gladstone & Cimpian, 2021; Zirkel, 2002).

1.3 Current study

Taken together, a growing body of literature suggests that low SES families shape adolescents' STEM motivation through several strengths such as socioemotional support, values-based messages, and exposure to familial role models. However, few studies have systematically examined how these distinct forms of influence in adolescence relate to STEM motivations among low SES youth who decided to pursue a STEM degree. As a result, our study aimed to investigate how low SES undergraduate students who are currently majoring in STEM felt their parents and families influenced their STEM pathways in adolescence.

To do this we examined qualitative open-response data to see which parental influences were most common for low SES STEM majors in their youth. Based on previous research (e.g., Aschbacher et al., 2010; Guiffrida, 2006; Packard et al., 2012) we predicted that socioemotional support, instrumental support, and role models would be commonly reported influences. To next address our key research question we wanted to investigate if commonly reported sources of influence (e.g., instrumental and socioemotional support) during adolescence would statistically predict current STEM majors' motivation. We hypothesized that perceived parent support recollected as occurring during adolescence would be positive predictors of STEM

motivation amongst current STEM majors from low SES backgrounds. In particular, based on past research (e.g., Starr et al., 2025), we anticipated that socioemotional support would be particularly related to STEM expectancy and value beliefs.

2 Method

2.1 Participants

Our sample included 222 undergraduate students (86% women, 12% men, 1% non-binary/enby). Undergraduates qualified for this study if they (1) were majoring in or planned to major in STEM (79% had declared a STEM major); and (2) self-identified as coming from a low-income background (73%) or as first-generation college students (66%); nearly half (40%) reported being both first-generation and from a low-income background. Six students did not answer the qualitative question and thus were included in the quantitative but not qualitative result sections. A majority of students identified as White (58%, $n = 128$), followed by Latine (19%, $n = 43$), Asian (17%, $n = 38$), Black (7%, $n = 18$), or Native American (2%, $n = 5$). (Participants could identify as multiple racial/ethnic identities; 3% did so.) Maternal education levels were as follows: less than a high school degree (12%, $n = 27$), high school degree (36%, $n = 77$), associate's degree (29%, $n = 62$), bachelor's degree (17%, $n = 37$), or more than a bachelor's degree (7%, $n = 14$). Additionally, 60% reported receiving a Pell grant to attend college.

Participants were on average 20 years old ($SD = 1.53$) and had been taking undergraduate courses for two years ($SD = 1.03$). First year students made up 18% of the sample, sophomores 44%, juniors 22%, and seniors 17%. Participants had on average taken nine STEM courses in college ($SD = 6.16$) and had a 3.33/4.0 overall GPA ($SD = 0.50$). Participants were most commonly majoring in a health science field (e.g., biology, pre-nursing). Specifically, 33% of the sample was majoring in pre-nursing ($n = 74$), 18% of the sample in neurobiology or neuroscience ($n = 39$) and 11% in biology ($n = 24$). Additionally, 12% were majoring in psychology ($n = 27$) and 7% in speech pathology or communication sciences and disorders ($n = 16$). For a full list of participant majors, see Supplementary Table S1.

2.2 Procedure

Participants were recruited through the educational psychology participant pool at a large midwestern university from spring semester 2024 through fall semester 2024. The educational participant pool includes a large neuroscience and education course that students from several STEM majors are required to take. Participants took the survey on Qualtrics during their own time in a location of their choosing. The survey opened with an Institutional Review Board (IRB) approved consent form from the first author's university, informing students that participation was voluntary. Students received course credit for their participation.

2.3 Positionality statement

The first author is a White woman from a low-income, low SES background. She was the first in her family to graduate from college. Her experiences growing up helped inspire her to study parent support among low SES families. The second author is a White woman from a middle-class background. Her parents obtained master's degrees offering both academic and financial capital to attend college and post baccalaureate degrees.

2.4 Measures

For a full list of items, see Supplementary Materials Table S2. For a correlation matrix of major demographic and quantitative variables, please see Table 1.

2.4.1 Demographic variables

First-generation and low-income status were asked using the following two questions: "Are you a first-generation college student? (i.e., the first in your immediate family to go to a four-year college)." and "Do you identify as coming from a low-income background? (some signs of growing up low-income are receiving a Pell grant for college, or free or reduced lunch in school)." Students answered either yes or no. Additionally, gender, race/ethnicity, and GPA were asked about and used as control variables in the regression analysis. To measure gender, participants were asked "What gender do you identify as? (check as many as apply)". They could choose from man, non-binary/gender queer, woman, or they could write in their own gender identity. To measure race/ethnicity, participants were asked "What race/ethnicity do you identify as? (check as many as apply)" They were given options as identifying as the major race/ethnicities in the US or could write in their own race/ethnicity. People who identified as Black, Latine, or Native American were categorized as having underrepresented minoritized (URM) status in STEM. People who identified as Asian, Middle Eastern, or White were categorized as not having URM status in STEM (non-URM). To measure GPA, participants were asked "What is

Table 1 Correlational table of major quantitative variables

Construct	1	2	3	4	5	6	7	8	9
1. First-generation	–								
2.Low-income	−0.43**	–							
3.Woman	−0.01	−0.01	–						
4.URM	0.13	0.19**	0.1	–					
5. GPA	−0.08	−0.12	0.14*	−0.20**	–				
6. Instrumental support	−0.14	−0.09	−0.03	0.04	0.14	–			
7. Socioemotional support	−0.22*	−0.02	0.06	0.00	0.20*	0.38**	–		
8. Expectancy beliefs	0.00	−0.09	−0.03	0.01	0.17*	0.24**	0.17	–	
9. Value beliefs	0.11	−0.04	0.15*	0.08	0.07	0.17	0.17	0.60**	–

*Correlation is significant at $p < .05$ level

**Correlation is significant at $p < .01$ level

your overall GPA?"; thus, GPA reflects a student's self-reported GPA at the time they participated in the study.

2.4.2 Qualitative measures

Participants were asked through an open-response question to share how their lived experiences influenced their choice to pursue STEM. Participants were asked, "Please tell me how your parents and home environment influenced your interest and persistence in STEM as a teenager".

2.4.3 Quantitative measures

2.4.3.1 Parent support Parent support during adolescence was quantitatively measured using four items which measured instrumental support and two items which measured socioemotional support. Both fell onto separate constructs in an exploratory factor analysis (see Supplementary Table S4). Initially, two items were also written to capture role models in STEM; however, factor analysis indicated that they did not fall into a separate component. Thus, these two items were excluded from the present analyses. Items were developed based on previous literature about how parents support their adolescents in STEM (e.g., Starr et al., 2022a, 2025) and based on other scales measuring parent STEM support (e.g., Lee & Simpkins, 2021; Simpkins et al., 2015; Wang & Sheikh-Khalil, 2014). Participants were instructed to answer whether their parents supported them in these ways during their adolescent years. An example item for instrumental support is: "My parents did STEM activities at home with me growing up" ($\alpha = 0.86$). An example item for socioemotional support is: "My parents encouraged me in STEM growing up" ($\alpha = 0.91$). Possible responses ranged from 1 (*strongly disagree*) to 6 (*strongly agree*).

2.4.3.2 Current STEM motivational beliefs Participants were asked about two kinds of STEM motivational beliefs: expectancy beliefs and value beliefs. Items were created based on situated expectancy-value theory (Eccles & Wigfield, 2020). Items have been used in past research (e.g., Hsieh et al., 2021; Jiang et al., 2020; Starr & Leaper, 2024). Expectancy beliefs were measured using four items (example item: "I am certain that I can master STEM skills") ($\alpha = 0.86$). Value beliefs were measured using four items which represented the four distinct categories of value beliefs (interest, attainment, utility, and cost). Interest item: "I find my STEM courses interesting and fun"; attainment item: "I value what I learn in my STEM courses"; utility item: "What I learn in my STEM courses is useful for life after college"; cost item: "being in STEM courses is worth the effort" ($\alpha = 0.83$). Possible responses ranged from 1 (*strongly disagree*) to 6 (*strongly agree*). Prior research using situated expectancy-value theory as a framework has similarly investigated these value belief categories as combined into one scale (e.g., Debatin et al., 2025; Perez et al., 2014; Tulagan & Eccles, 2023). Although some studies find that the three value beliefs are factorially distinct, they are typically highly correlated (Wigfield & Eccles, 2020). For the present project we elected to combine them into one scale given prior research that

suggests the items can be combined. Additionally, our EFA suggested they fit onto the same dimension (factor loadings = 0.73–0.81).

2.5 Data analysis

2.5.1 Qualitative data analysis

Using an inductive approach (Braun & Clarke, 2006), the open-response questions were reviewed, and corresponding thematic maps were created. Seven coding categories of parental influence were created as highlighted through the inductive map process: socioemotional support, instrumental support, role models, STEM for stability, STEM family values, challenging home environment, and discouragement. Because this paper is focused on parent support, we do not report discouragement or challenging environment results in the present paper. The authors coded 20% of the sample to obtain inter-rater reliability (87% agreement, $k = 0.83$). Disagreements in codes within the reliability set were discussed to ensure the authors remained aligned in the coding of the remaining sample. The description and an accompanying example of each coding category as well as frequency of endorsement is included in Table 1.

2.5.2 Quantitative data analysis

Analyses were conducted using IBM's Statistical Package for the Social Sciences (SPSS) Version 31. To test our hypothesis that socioemotional and instrumental support (quantitatively measured) would positively relate to current STEM expectancy and value beliefs even after controlling for GPA, gender, and race/ethnicity, we ran four 2-step regressions with expectancy beliefs as the outcome for two models, and value beliefs as the outcome for the remaining two models. The covariates GPA, gender (0 = *man*, 1 = *woman*), and underrepresented minority status (URM) (0 = *non-URM*, 1 = *URM*) were entered in step 1 Socioemotional or instrumental support were then added in step 2. Most participants had no missing data. All but three participants completed the quantitative items. Two people did not answer the second socioemotional support item ("My parents encouraged me to pursue STEM jobs growing up"); however, both participants had answered the other STEM emotional support item and thus were included in the analyses using their response to that item. Additionally, one participant did not answer the value beliefs question "Being in STEM courses is worth the effort", however they answered the three other value belief items; thus, a mean of these three items was used.

For descriptive purposes, we additionally examined the qualitative experiences of participants. We first examined the patterns of responses within our coding scheme descriptively by calculating the proportion of participant responses that included mention of a particular code. Second, to examine whether the presence of specific content codes varied by student background, we conducted a series of chi-square tests of independence. For each of the five coding categories related to parent support, binary indicators of presence vs. absence of the code were cross tabulated with student group membership (first-generation only, low-income only, and both first-

generation and low-income). Pearson's chi-square tests were used to assess whether the distribution of code presence differed across groups. All tests were conducted using a significance threshold of $\alpha = 0.05$ with Bonferroni adjustments.

3 Results

3.1 Descriptive findings

Table 1 provides a correlation matrix of major demographic and quantitative variables. Additionally, Supplementary Table S4 provides qualitative coding proportions for the sample and Supplementary Table S5 provides means and standard deviations for all major quantitative variables. Both tables provide descriptive statistics for the total sample and separated by participants who are both first-generation and low-income, first-generation only, and low-income only. Regarding the qualitative measures, chi-square tests indicated that students who were both first-generation and from a low-income background reported significantly less instrumental support compared to students who were low-income or first-generation only. Regarding the quantitative measures, ANOVAs with follow-up Tukey tests indicated that students who were low-income only reported significantly higher socioemotional support compared to students who were both low-income and first-generation.

3.2 Qualitative findings

To examine the qualitative experiences of participants, we first examined the patterns of responses within our coding scheme (see Table 2 for example quotes and percentages of responses that included each type of influence). Looking descriptively at the data, socioemotional support through encouragement in STEM was the most common source of family influence, with over one third of respondents (35%) highlighting this as an important motivator in their STEM pathway. Socioemotional support from participants' families mirrored expectancy-value patterns. Participants shared that their families helped support them through recognizing and celebrating their STEM abilities and interests. For example, a low-income student shared, "My parents influenced me by noticing my strengths and pointing them out to me. I was always much better in science-based courses throughout school, so they encouraged me to pursue interests in that area." Similarly another low-income participant shared that their parents supported their pathway in STEM through recognizing their strength in math. They said, "My parents encouraged me to take harder mathematics classes at my school because I scored above average on school state exams. I enjoyed math and my parents helped me with these harder courses if I needed it." Other responses focused on the encouragement families provided through recognizing STEM-related interests. For example a participant who identified as the first in their family to go to college shared, "My parent and home environment influenced my interest in STEM because they always encouraged me to explore my interests. They were both very supportive of me going into my interested career and were always very encouraging of me" Similarly a participant from a low-income background highlighted, "They

Table 2 Parental support qualitative codes: percentage agreement, descriptions, and examples

Measure	Percent agreement	Definition	Example	%, *n* endorsed
Socio-emotional support	75	Respondent received emotional forms of support through encouragement to pursue/engage with STEM.	"My family encouraged me to pursue my passions and sharing with them that I was interested in STEM always led with support rather than discouragement."	35%, *n* = 75
Instrumental support	77	Respondent received support in STEM through coactivity, advice giving, provision of materials/activities, or support on applications/homework.	"My dad always took me to science fairs and helped me garner an interest in mathematics."	28%, *n* = 61
Role model	90	Respondent experienced exposure to STEM fields due to a parent, sibling, or other adult's occupation/educational background.	"[My mom] works in healthcare as a radiology technician. Watching her work through healthcare and joining her to work some days has influenced me to go into healthcare."	23%, *n* = 50
STEM as Stability	100	Respondents mentioned the financial value/social mobility of a career in STEM was communicated by their parents.	"My parents were more supportive toward career routes that were STEM-related as they believed and therefore emphasized pursuing a career in STEM to secure a stable and financially rewarding future."	15%, *n* = 32
STEM Family Value	100	Respondent described how their parents values supported their interest or dedication to STEM. Values included ideas about the importance, learning, curiosity, or serving a bigger purpose (e.g. helping the community).	"From an early age, my parents fostered a sense of curiosity and wonder about the world around me. They encouraged me to ask questions, explore new ideas, and seek out answers through experimentation and research."	14%, *n* = 31

supported me no matter what I wanted to do. They encouraged me to find what I truly enjoyed, and STEM was definitely something I am passionate about."

The second most common influence was instrumental support (28%) where respondents shared that their parents or families motivated them through the provision of STEM-related materials, access to STEM spaces, or STEM mentorship. Several participants shared that their parents supported them in getting STEM-related materials like books or videos. Others shared that they received support to access STEM spaces, like after school programs, science clubs, science fairs, or STEM camps. For example, a first-generation and low-income student shared, "They [my parents] also encouraged participation in STEM activities such as after school clubs or camps." Other participants shared that their families helped them gain access to STEM professionals through job shadowing or other forms of mentorship. One low-income student shared, "She [my mother] took me with her to many of her doctor appointments and even got me a chance to shadow her ENT when I was in high school." Another student who was from a first-generation background shared how his parents not only provided emotional encouragement but also supported him in learning more about STEM. He said, "They were both very supportive of me going into my interested career and were always very encouraging of me that I do job shadows to help me understand even more what my interests are. They also provided me with resources and helped me along the way."

About one in four (23%) participants shared that a parent or close family member served as a role model for them in a STEM career which played a key role in shaping their STEM motivation. For example, one first-generation student from a low-income background reported that she had received socioemotional support in addition to her mother being a role model and her parents valuing STEM as a way to achieve economic mobility: "With my mother being a medical assistant I was always exposed to medical devices that she would use. Growing up I had a big passion for art and while my family definitely encouraged it as a hobby, as a career they knew that coming from our economic background that it wouldn't be a sustainable one. As I grew up and spent more time in the hospital whether that be in it for a personal reason or just stopping by to see my mom I grew I love for the space. Once my mom found out that I wanted to be a surgeon she definitely pushed me towards it but once I realized that nursing was the route I wanted to take she was beyond encouraging."

While many respondents shared that their immediate family members such as parents impacted their STEM pathways, several participants also shared that extended family served as meaningful role models as well. For example, a student who was the first in his family to attend college shared, "One factor that did help me gain interest in STEM was my uncle while I was growing up he was in high school and eventually he enlisted in the Air Force to pursue working on jets. Whenever I got to see him he always tried to take me to museums, constantly explain STEM related topics or do activities that I now realize started an interest in the STEM field. Lastly I really looked up to him so ever since I was little I wanted to follow in his footsteps and be successful just like him."

In addition to support and role models, participants also highlighted their families' views of STEM were influential. These views included two common pathways. The first was that families had promoted STEM values (14%) related to the importance of

lifelong learning, curiosity, or helping others. For example, a low-income first-generation student shared, "My father always encouraged me and my siblings to be curious about learning especially in scientific aspects". Another first-generation participant shared, "Even that [though] I am the first-generation college student at my home, I think I received a lot of support or at least interest from my father. He was interested in biology but did not finish after high school, but he didn't stop learning on himself through going to lectures or listening to seminars online."

The other pathway that participants shared relating to values was that their parents promoted STEM as an opportunity to have stability in their lives, particularly by doing well for themselves financially and/or promoting their social mobility. One participant who was a low-income student shared, "I wouldn't say my parents played a major role in my interest in STEM, but they did keep my interest alive and motivated me to pursue my dreams. As children of immigrants with low-income, they want nothing but to make sure their children end up in very successful and fulfilling careers and, of course, with lots of money. So, they were thrilled that I only needed a little convincing to pick a career in STEM, specifically medicine." Similarly, one woman who was both first-generation and low-income shared, "Growing up in a low-income household, I knew that whatever I was going to study it should be a career that will let me make a living. My parents encouraged me when I told them I was going to go into the medical field. They told me to succeed whatever they couldn't succeed when they were younger."

The themes revealed within the open responses of current STEM undergraduates highlight the robust positive influence parents and other family members have on the STEM pathways of adolescents from low SES backgrounds. From these themes, socioemotional support and instrumental support were the most common influences. As a result, we investigated how quantitative methods of these types of support (socioemotional and instrumental) predict current STEM motivational beliefs.

3.3 Hierarchical regression findings: socioemotional and instrumental parent support as predictors of current STEM motivational beliefs

As expected, though our hierarchical regression analyses we found that parent socioemotional support positively related to students' current STEM expectancy beliefs and value beliefs above and beyond GPA and demographic variables (see Tables 3 and 4). However, parent instrumental support significantly and positively related to current STEM expectancy beliefs, but not value beliefs (see Tables 5 and 6).

4 Discussion

This study sought to investigate how parent and familial influences promote STEM pathways among low SES students from adolescence to their current motivations as STEM majors in college. The vast majority of participants (84%) qualitatively reported that their parents or family had supported them in STEM as an adolescent in some way. According to the open responses of students currently majoring in STEM, socioemotional and instrumental support were the two most common parental sup-

Table 3 Regression coefficients of socioemotional parent support on STEM expectancy beliefs

STEM Expectancy Beliefs						
Variable	Model 1			Model 2		
	B	SE	Beta	B	SE	Beta
Constant	3.32***	0.55		2.93***	0.56	
GPA	0.40*	0.16	0.24	0.33*	0.16	0.19
Woman	−0.43	0.26	−0.16	−0.45	0.25	−0.16
URM	0.07	0.19	0.04	0.06	0.18	0.03
Socioemotional support				0.14*	0.06	0.21
Adj. R^2			0.04			0.07
$R^2\Delta$						0.04*

URM = underrepresented minoritized race/ethnicity in STEM

*p <.05 ***p <.001

Table 4 Regression coefficients of socioemotional parent support on STEM value beliefs

STEM Value Beliefs						
Variable	Model 1			Model 2		
	B	SE	Beta	B	SE	Beta
Constant	4.28***	0.43		4.02***	0.45	
GPA	0.16	0.13	0.13	0.11	0.13	0.09
Woman	0.14	0.21	0.07	0.13	0.20	0.06
URM	0.06	0.15	0.04	0.05	0.15	0.03
Socioemotional support				0.11*	0.05	0.19
Adj. R^2			0.00			0.04
$R^2\Delta$						0.04*

URM = underrepresented minoritized race/ethnicity in STEM

*p <.05 ***p <.001

Table 5 Regression coefficients of instrumental parent support on STEM expectancy beliefs

STEM expectancy beliefs						
Variable	Model 1			Model 2		
	B	SE	Beta	B	SE	Beta
Constant	3.32***	0.55		2.99***	0.55	
GPA	0.40**	0.16	0.24	0.34*	0.16	0.20
Woman	−0.43	0.26	−0.16	−0.38	0.25	−0.14
URM	0.07	0.19	0.04	0.05	0.18	0.02
Instrumental support				0.16**	0.06	0.23
Adj. R^2			0.04			0.08
$R^2\Delta$						0.05**

URM = underrepresented minoritized race/ethnicity in STEM

*p <.05 **p <.01. ***p <.001

ports of their STEM pathways. STEM role-models, family values related to STEM, and communicating STEM as a path to financial stability, were also common supports of STEM pathways during adolescence, but not as common as socioemotional and instrumental support. Hierarchical regression analyses demonstrated that both

Table 6 Regression coefficients of instrumental parent support on STEM value beliefs

STEM value beliefs						
Variable	Model 1			Model 2		
	B	SE	Beta	B	SE	Beta
Constant	4.28***	0.43		4.11***	0.44	
GPA	0.16	0.13	0.13	0.13	0.13	0.10
Woman	0.14	0.21	0.07	0.17	0.20	0.08
URM	0.06	0.15	0.04	0.05	0.15	0.03
Instrumental support				0.08	0.05	0.16
Adj. R^2			0.00			0.01
$R^2\Delta$						0.03

URM = underrepresented minoritized race/ethnicity in STEM

***$p < .001$

socioemotional support (e.g., encouragement) and instrumental support as recalled during adolescence were strong predictors of students' current STEM motivations. A strong body of prior work and theoretical grounding through situated expectancy-value theory has demonstrated that parent support is a key predictor of one's expectancy beliefs (e.g., can I do the task) and value (e.g., do I want to do the task) (Hill & Tyson, 2009; Simpkins et al., 2015, 2018; Tulagan et al., 2022). Our study confirmed that this is also the case for STEM students who come from low SES backgrounds.

Analyses also indicated some differences by student SES background, such that students who were both first-generation and low-income were less likely to receive instrumental and socioemotional support. This illustrates how the intersections of marginalized identities can lead to multiple forms of disadvantage and compound negatively on youth outcomes. Families who have less financial, social, and academic capital may find it more difficult to support their child in STEM, especially in more costly methods of support such as instrumental support. Despite this, a majority of students (72%) who were both first-generation and from low-income backgrounds reported receiving some kind of parental support. These findings suggest that parent support in STEM is important for later motivational outcomes among low SES families. Below we further review these findings through their connections and contributions to the existing body of literature.

4.1 Socioemotional support and instrumental STEM support

Socioemotional support was the most common form of parent support reported among participants. Importantly, we found socioemotional support to predict both expectancy and value beliefs in STEM. This aligns with past research that indicates that socioemotional support, such as encouragement in STEM, may be particularly predictive of marginalized student's motivational beliefs in STEM (e.g., Starr et al., 2025). Our study suggests that parental socioemotional support—such as encouragement and high expectations—can significantly boost adolescents' motivation and achievement, including in STEM. These messages that affirm competence, express pride, and encourage persistence help adolescents develop confidence and value for academic subjects.

We found that instrumental support was the second most common reported influence in low SES students' STEM pathways. Students in our sample frequently reported that their parents engaged them in traditional ways of instrumental support, such as participating in science fairs or attending afterschool STEM programs. This somewhat contradicts previous work which has framed students from low SES backgrounds as lacking instrumental support from their families due to barriers in academic and social capital (e.g., Starr et al., 2022a, 2022b). Furthermore, we found that the instrumental support students recalled receiving in adolescence was predictive of their current STEM expectancy beliefs. These findings mirror prior work documenting links between parent coactivity (a form of instrumental support) and student competence (Simpkins et al., 2005) but expand upon the literature to reveal that these patterns are also present amongst students from low SES backgrounds.

Much of the existing prior literature about parent STEM support among middle class and low SES families has examined general support as one category (e.g., Starr et al., 2022b), overlooking that different kinds of parent support, such as socioemotional support and instrumental support, may differ. Our study adds to this research by considering both kinds of support, finding that both related to motivational beliefs in slightly different ways. The finding that socioemotional support was more predictive of STEM motivational beliefs than instrumental support aligns with past research on the importance of encouragement (e.g., Simpkins et al., 2012). This further mirrors previous research with low SES adolescents which found parent encouragement to be particularly predictive of adolescent's motivational beliefs relative to various forms of instrumental support, including advice giving and coactivity (Starr et al., 2025). Socioemotional support from parents may be particularly valuable for low SES adolescents given that they may not experience similar kinds of support in school (e.g., Archer et al., 2012; Mello, 2024). In contrast, instrumental support is potentially important for low SES students' expectancy beliefs because many forms of instrumental support, such as coactivity, directly target STEM skill-building (Estrada et al., 2018; Shirefley, 2025).

4.2 Low SES students leverage parent support as cultural strengths

This study highlights that low SES parents support their children in multiple ways in addition to instrumental and socioemotional support. One such strength amongst families was the direct messaging about the value of STEM pathways for social mobility. This aligns with prior work documenting that for many low SES families STEM careers are viewed as pragmatic ways to achieve financial stability and social mobility (Archer et al., 2012; Aschbacher et al., 2010). The responses from students within this sample highlight that these explicit messages play an important role in the development of youth's motivational beliefs about STEM. Additionally, our study revealed that parental messaging about the value of STEM was not isolated to finances or social mobility but was also linked to an importance in learning about how and why the world works the way it does. The communication of STEM as a topic of interest amongst low SES families may be an excellent motivational bridge for students to use in conjunction with other influences like general encouragement.

In tandem, these values may promote overall motivation and protect against potential barriers.

Lastly, student open responses within this study confirmed prior work highlighting that families promote student STEM pathways through role modeling (Delgado et al., 2021; Gladstone & Cimpian, 2021; Zirkel, 2002). For example, nearly a quarter of respondents highlighted that as youth they were exposed to STEM through family members who worked in STEM fields. Interestingly, most of these STEM careers were what are considered "blue-collar" or skilled technical workforce fields (e.g., medical assistants, mechanics, agricultural workers). So, while these youth potentially lacked academic or social capital to traditional STEM domains (e.g., engineers, computer scientists), youth were building rich community-centered connections within the STEM fields around them. These findings support existing literature documenting that STEM pathways are strengthened through connections to existing funds of knowledge (Moll et al., 1992) and meaningful role models (see Gladstone & Cimpian, 2021 for a review). Notably, there are many STEM careers that do not require a 4-year college degree, including working as a skilled machinist or as a nurse assistant (McCrary, 2017). As a result, these findings offer insight into ways in which STEM pathways can be more strongly promoted through programming that recognizes the existing STEM-related capital low SES youth have through their family and community connections. Future studies might examine role models in these blue collar STEM jobs as a way to spark interest, value, and perception of abilities in STEM among low SES youth.

4.3 Practical implications

As noted above, our findings have several implications for parents and educators supporting low SES students in STEM. First, low SES parents and families are an important source of support in STEM for adolescents due to strong socioemotional support, instrumental support, STEM values, and role modeling. Educators might build on these kinds of support by informing parents that socioemotional support through encouragement is very important and can help students build both strong competence beliefs as well as value in STEM. Additionally, parents should know that when they promote the importance of pursuing STEM to help others, learn about the world, or for financial stability this might be particularly motivating for students (for an intervention that does this among middle-class families, see Harackiewicz et al., 2012).

Educators may also need to reflect on the STEM strengths their students have and check potential biases. Within the current sample, instrumental support was reported as the second-most common influence which counters existing stereotypes, and some research, that youth from low SES backgrounds do not have families with knowledge or access to STEM to help their children. Parents or other close family members may be able to help their children outside of school, especially when educators work in collaboration with families and recognize both areas of content knowledge strength (e.g., blue-collar STEM) and areas that youth may need additional support.

Lastly, to build on existing strengths, educators may reflect on the types of STEM they are promoting within their science-learning opportunities and find ways to

incorporate culturally relevant examples. Given that role models were important influences for students within this study, educators may seek to incorporate STEM careers represented within the workforce of youths' parents and extended families. Educators may ask students to reflect on ways their parents or members of their community use STEM skills in their own jobs (e.g., as a construction worker) or by reflecting on ways in which STEM skills are or could be used to help others (e.g., trauma-informed nurses could help navigate the opioid epidemic). By reflecting on their environment, students may boost their value beliefs as they see STEM skills are used in many occupations and have real-world applications for their futures.

4.4 Future directions and limitations

Although this study provides a unique perspective on parental influence in STEM among low SES families, it is not without limitations. First, the study is retrospective, and participants were currently pursuing STEM majors in college. Although this allows us to understand past support among low SES students who have had some success at pursuing STEM, it leaves out low SES students who were not able to pursue STEM. Thus, our study may overestimate parent influence and overlook barriers. Additionally, our study was correlational, and directionality cannot be assumed. It is possible that students who currently have stronger motivational beliefs in STEM may be more likely to recall being supported by their parents. Future studies might examine parent support or influence in STEM among low SES adolescents longitudinally.

Second, this study asked participants to answer a set of questions in an online survey, rather than in an interview or focus group. Although this allowed us to collect a large amount of data and allowed students the privacy to reflect upon and write out their answers, it meant that there were no follow-up questions from an interviewer. Future studies might interview a smaller number of participants one on one or in a focus group to dive deeper into the lived experiences of students from low SES backgrounds to capture greater complexity (e.g., do participants feel that the socioemotional support they received mattered more than instrumental support?). Furthermore, we identified low-income participants using a binary measure which required students to self-identify as low-income. People in the U.S. have the tendency to over-identify as middle class even when their family income suggests they are not (Samuel, 2013), although this may be changing (Brenan, 2024). As a result, we may have missed some students that are classified as low-income based on their family's income but do not identify as such.

Third, we acknowledge that our value beliefs measure may not fully capture situated expectancy-value theory. We measured STEM expectancy beliefs and value beliefs separately, but did not separately measure the value sub-components of interest, utility, cost, and attainment value (Eccles & Wigfield, 2020). Relatedly, we did not have a quantitative measure of STEM family values or STEM for stability, both of which emerged in our qualitative findings. While we included a measure of STEM role models, it did not sufficiently differentiate in our exploratory factor analysis. Future studies should develop additional measures to examine these forms of parent support quantitatively as they appear to matter qualitatively for youths' STEM pathways.

Finally, women made up 86% of our sample; thus, our findings may primarily reflect gender-specific experiences and may not be fully representative of other genders. Many STEM fields have large gender discrepancies; for example, 74% of people who work in health-related jobs are women, while only 15% of people in engineering jobs are women (Pew Research Center, 2021). Furthermore, the environment and experiences of women in health-related vs. non-health related STEM fields differ, with women in fields such as computer science and engineering frequently experiencing more gender discrimination and a chilly climate (e.g., Lee & McCabe, 2021; Walton et al., 2015). Aligned with this, a majority of our sample was majoring in health science fields, such as nursing. Low SES women may experience different kinds of parent support regarding these health-related fields compared to more male-dominated STEM fields. For example, health-related fields may be more likely to be perceived as affording helping goals and aligning with female-gender norms compared to engineering and computer science (Diekman et al., 2015, 2016; Leaper, 2015). Future studies might purposely recruit low SES participants from health and non-health related STEM fields and compare the kinds of parent support students experienced by gender.

4.5 Conclusion

Taken together, this mixed-methods study demonstrates that low SES families' socioemotional support, instrumental support, values, and role models are key in shaping adolescents' STEM pathways that matter for their later STEM motivations. Across both qualitative and quantitative evidence, we found that socioemotional and instrumental support were the two most common influences, and both mattered for students' current STEM expectancy-value beliefs. However, students also noted family values and role models were key motivators of their STEM pathways within adolescence. These findings extend expectancy-value theory by illustrating how motivational beliefs are socialized within different contexts that leverage cultural funds of knowledge and strengths. This work highlights the importance of family-centered approaches to STEM outreach that promote families to continue encouraging their children in STEM while leveraging their family values and cultural connections through role models in STEM careers.

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Declarations

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Ethical approval Ethical considerations for this Human Subjects research were reviewed and deemed acceptable by the University of Wisconsin-Madison's Institutional Review Board. Informed consent from participants was obtained prior to data collection.

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